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1. Introduction

This document describes the `payments` architecture using authored functional design and inferred runtime/code realization. It is intended to help product, `platform`, security, and implementation engineers reason about one shared model. Functional design is kept stable while realization details are inferred from infrastructure and source artifacts.

1.1. Scope and Assumptions

- `Functional Groups` and `Functional Units` are the stable authored design backbone.
- `Runtime elements` and `code elements` are inferred from available IaC, deployment, and source artifacts.
- Inferred coverage is evidence-driven and may be partial when source metadata is missing.
- `Requirements` and reference indexes are provided for traceability, not as primary narrative flow.

2. How To Read This Document

This document is intentionally split into two layers:

1. Concept layer: functional and view-specific architecture narratives for human understanding.
2. Evidence layer: generated requirement traceability and inferred reference indexes.

Read in this order:

1. Architecture Intent View: stable authored architecture intent and ownership boundaries.
2. Communication / Deployment / Security Views: viewpoint-specific operational behavior.
3. Traceability View: requirement mapping, evidence coverage, and confidence.
4. Reference Appendix: supporting reference IDs and inferred-source links.

3. Document Health Snapshot

View	Authored Status	Authored Status Explanation	Units in Scope	Direct Evidence Coverage	Gap Count	Heuristic Confidence	Heuristic Basis
Architecture Intent View (`VIEW-ARCHITECTURE-INTENT`)	stable	Stable authored capability boundaries and ownership split; changes are intentional and infrequent.	6	6	0	high	Defaults authored units to covered; use authoredStatus as the primary publication signal.
Communication View (`VIEW-COMMUNICATION`)	in-review	Communication semantics are partially inferred and still being refined with explicit protocol and interaction metadata.	4	3	1	medium	Counts a unit as covered when direct inferred runtime evidence exists.
Deployment View (`VIEW-DEPLOYMENT`)	in-review	Deployment chain and control-plane mapping are mostly present, but ownership evidence remains incomplete for some units.	6	5	1	high	Counts a unit as covered when direct inferred runtime evidence exists.
Security View (`VIEW-SECURITY`)	draft	Threat mapping is present for key vectors, but control expectations and coverage breadth are still evolving.	6	6	2	high	Counts a unit as covered when explicit authored attack-vector mapping exists.
Traceability View (`VIEW-TRACEABILITY`)	in-review	Requirement-to-unit linkage is present; cross-layer ownership consistency still needs tightening.	6	5	4	high	Counts a unit as covered when direct inferred runtime or code evidence exists.

4. Terms and Definitions

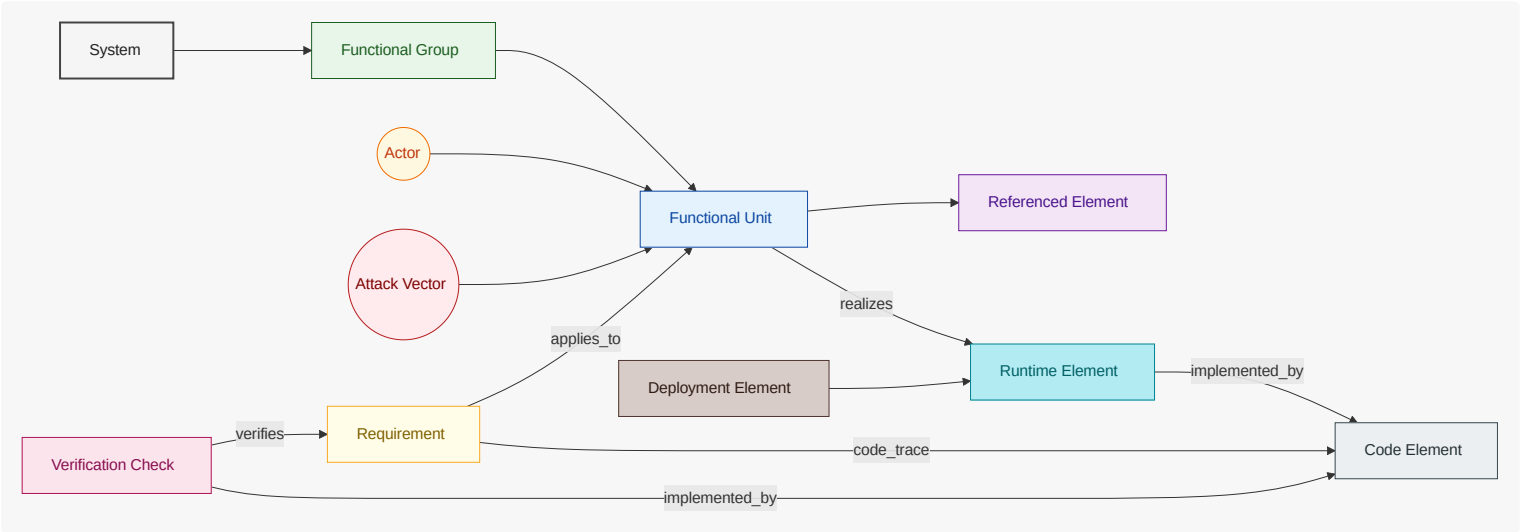
Term	Definition
`3DS verification is required` (COND-3DS-REQUIRED)	Condition indicating strong customer authentication must be performed. Aliases:3DS required
`actor` (EM-ACTOR)	A person or operational role that interacts with functional units.
`attack vector` (EM-ATTACK-VECTOR)	A technical misuse or attack path that targets functional, referenced, or runtime elements.
`authored mapping` (EM-AUTHORED-MAPPING)	An explicit relationship declared in architecture inputs between authored or referenced elements.
`bank gateway endpoint` (REF-BANK-GATEWAY-ENDPOINT)	External banking endpoint used for card authorization and decline decisions. Aliases:bank API
`bank link is available` (STATE-BANK-LINK-UP)	Connectivity state showing bank gateway communication is currently healthy. Aliases:bank link up
`bank link is unavailable` (EVT-BANK-LINK-UNAVAILABLE)	Event raised when payment engine cannot reach the external bank gateway. Aliases:bank link down
`checkout handling` (FU-CHECKOUT)	Handles checkout session start and customer-facing payment messaging. Aliases:checkout
`checkout is active` (STATE-CHECKOUT-ACTIVE)	Processing state where the user is actively submitting or confirming payment. Aliases:checkout active
`cluster provisioning` (FU-CLUSTER-PROVISIONING)	Provides cluster-level platform setup and environment bootstrap. Aliases:day1 provisioning
`code element` (EM-CODE-ELEMENT)	An inferred code structure or ownership element discovered from source trees and build metadata.

Term	Definition
compromised dependency (AV-COMPROMISED-DEPENDENCY)	Supply-chain compromise in runtime image or code dependency. Aliases:dependency compromise
customer (ACT-CUSTOMER)	End user submitting payment details during checkout. Aliases:buyer
decline reason (DATA-DECLINE-REASON)	Structured reason code or message explaining why authorization failed. Aliases:decline code
fraud check is enabled (FEAT-FRAUD-CHECK)	Feature flag controlling whether risk scoring is required before authorization. Aliases:fraud check
fraud evaluation (FG-FRAUD)	Functional group for risk scoring and fraud review capabilities. Aliases:fraud detection system, fraud domain
fraudulent transaction pattern (AV-FRAUDULENT-TRANSACTION-PATTERN)	Coordinated abuse pattern using synthetic identities, stolen instruments, or velocity anomalies to obtain unauthorized approvals. Aliases:fraud abuse, payment fraud
functional group (EM-FUNCTIONAL-GROUP)	A major authored capability area that groups related functional units.
functional unit (EM-FUNCTIONAL-UNIT)	An authored working unit inside a functional group that owns specific behavior.
gitops operations (FU-GITOPS-OPERATIONS)	Operates declarative delivery and release management flows. Aliases:day2 operations
helm platform (REF-HELM-PLATFORM)	Runtime packaging and deployment substrate for workloads. Aliases:helm
inferred mapping (EM-INFERRED-MAPPING)	A discovered relationship that links inferred runtime/code elements upward to authored design.
malicious api request (AV-MALICIOUS-API-REQUEST)	Crafted payload attempting to bypass validation or authorization logic. Aliases:malicious request
normal mode is active (MODE-NORMAL)	Standard operating mode with full external dependency usage enabled. Aliases:normal mode
payment authorization (FU-PAYMENT-AUTHORIZATION)	Coordinates authorization flow and bank gateway interaction. Aliases:auth
payment authorization is requested (EVT-PAYMENT-AUTH-REQUESTED)	Event raised when checkout asks payment engine to authorize a transaction. Aliases:payment auth requested
payment id (DATA-PAYMENT-ID)	Stable identifier used to correlate a payment across services and records.
payment is declined (EVT-PAYMENT-DECLINED)	Event indicating authorization was rejected and customer decline handling must execute. Aliases:payment declined
payment review is pending (STATE-REVIEW-PENDING)	Transaction state requiring manual support review before final disposition. Aliases:review pending
payments (FG-PAYMENTS)	Functional group for checkout and authorization capabilities. Aliases:payments domain, payments system
platform (FG-PLATFORM)	Functional group for provisioning and platform operations capabilities. Aliases:platform engineering
platform operator (ACT-PLATFORM-OPERATOR)	Role responsible for GitOps and platform lifecycle operations. Aliases:platform ops
postgresql client driver (REF-POSTGRES-DRIVER)	Third-party library used for application data persistence connectivity. Aliases:postgres driver
referenced element (EM-REFERENCED-ELEMENT)	An architecture-relevant external, platform, or third-party dependency represented by role.
replayed authorization callback (AV-REPLAYED-AUTH-CALLBACK)	Replay of a valid callback to induce duplicate or unauthorized state transitions. Aliases:callback replay
requirement (EM-REQUIREMENT)	A structured requirement statement that defines expected system behavior and maps to functional ownership.
risk score (DATA-RISK-SCORE)	Numeric fraud score produced by risk evaluation logic.
risk score is high (EVT-RISK-HIGH)	Event emitted when computed fraud risk exceeds the configured high-risk threshold. Aliases:high risk
risk scoring (FU-RISK-SCORING)	Evaluates transaction risk and returns score classification. Aliases:fraud scoring
risk threshold is exceeded (COND-RISK-THRESHOLD-EXCEEDED)	Condition indicating risk score crossed a policy threshold for escalation. Aliases:risk above threshold
runtime element (EM-RUNTIME-ELEMENT)	An inferred runtime realization element discovered from infrastructure and deployment sources.
support agent (ACT-SUPPORT)	Operations role that investigates declines and high-risk review cases. Aliases:support
support agent is notified (EVT-SUPPORT-NOTIFIED)	Event indicating manual review notification was sent to support operations. Aliases:support notified
support review (FU-SUPPORT-REVIEW)	Supports manual review and escalation handling for risky transactions. Aliases:manual review
upward linking (EM-UPWARD-LINKING)	Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.

Term	Definition
verification check (EM-VERIFICATION-CHECK)	An inferred or authored verification artifact that validates requirement behavior with test evidence.

4.1. Diagram Color and Shape Legend

This legend applies to all Mermaid diagrams in this document.



5. Architecture Intent View

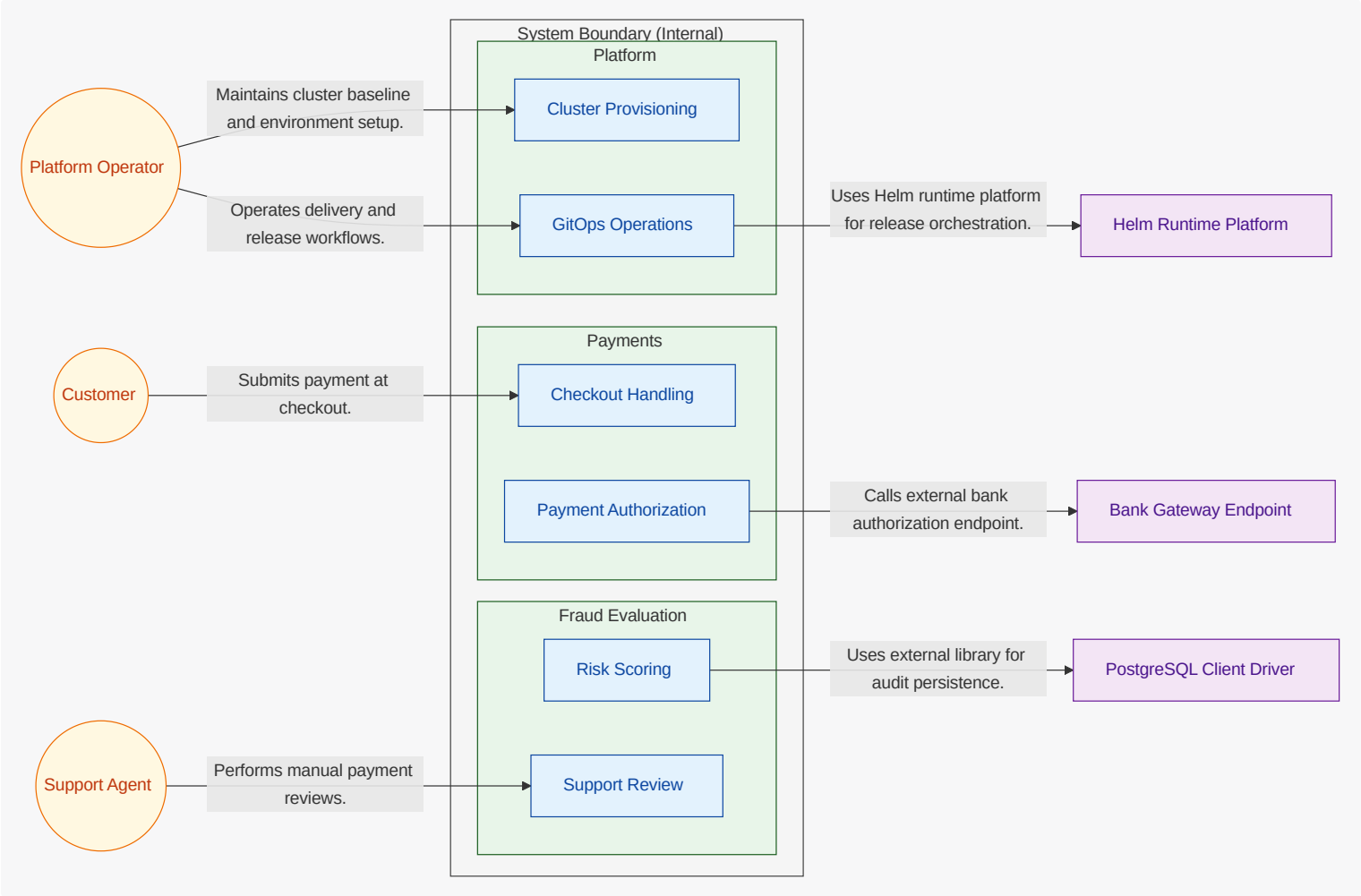
Show the stable authored architecture intent: capability areas, unit ownership boundaries, and intentional responsibility split.

5.1. What This View Answers

- What are the main capability areas?
- What are the stable units of responsibility and what does each unit own?
- How are responsibilities intentionally split?

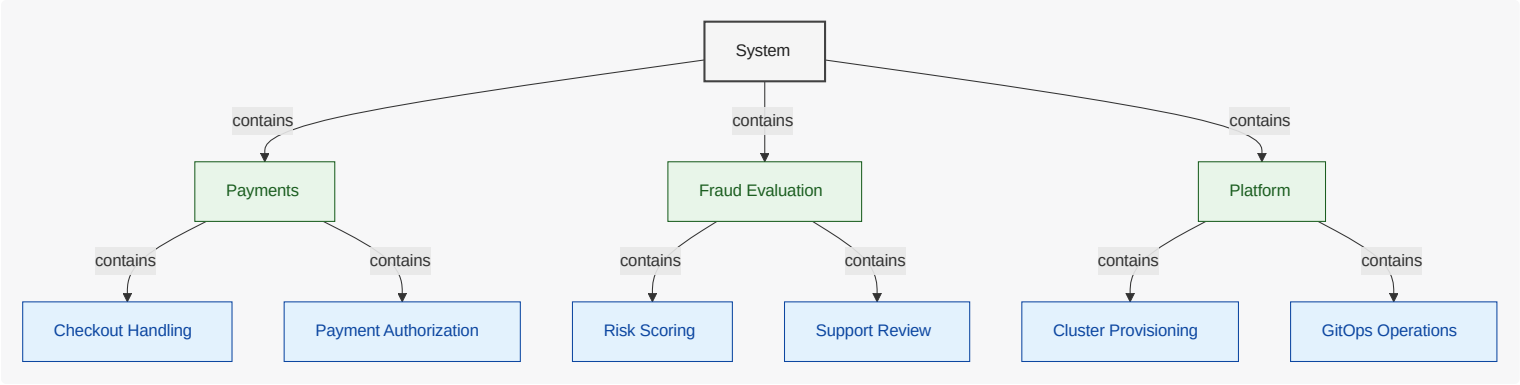
5.2. System Boundary Diagram

What this diagram shows: explicit internal vs external scope. Functional units are inside the `System Boundary (Internal)` container, while actors and referenced dependencies are shown outside the boundary.



5.3. System Decomposition Diagram

What this diagram shows: stable capability decomposition from functional groups into functional units.



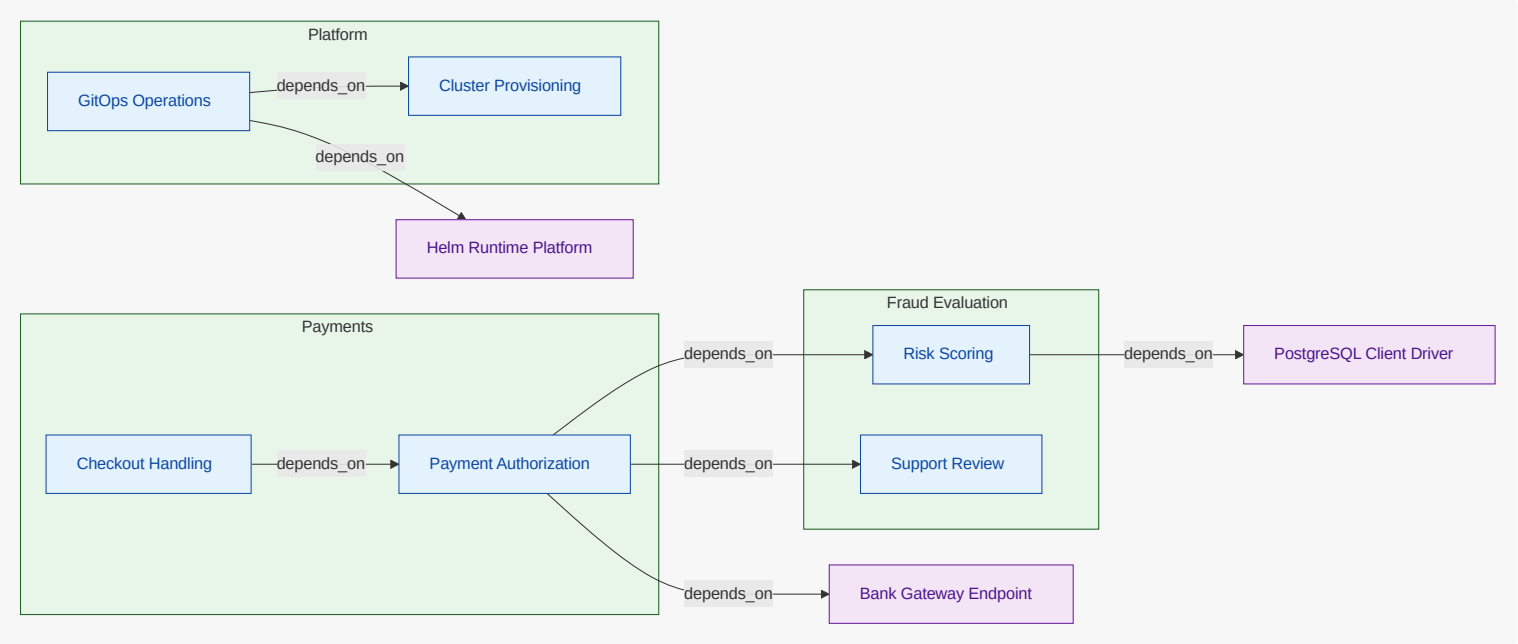
5.4. Functional Group Manhattan Matrix

What this diagram shows: a table-like matrix where functional groups are the bottom row (columns) and functional units are arranged in rows above their owning group.

Checkout Handling	Risk Scoring	Cluster Provisioning
Payment Authorization	Support Review	GitOps Operations
Payments	Fraud Evaluation	Platform

5.5. Functional Unit Dependency Diagram

What this diagram shows: dependency paths between functional units (grouped in Functional Group containers) and key referenced elements. Arrow meaning: `A -→|depends_on| B` means unit `A` requires behavior or data from `B` to fulfill its responsibility.



5.6. Functional Groups and Units (View Scoped)

5.6.1. Payments (FG-PAYMENTS)

Core payment `checkout` and authorization domain.

The `payments` group defines the product-facing payment journey from user intent to final payment outcome. It owns the consistency of `customer` messaging and transaction decisions when a payment is attempted, declined, retried, or temporarily unavailable. The group keeps `customer` flow stable while depending on fraud and external bank capabilities.

Checkout Handling (FU-CHECKOUT)

Checkout handling is the user-facing entrypoint where payment requests are initiated and normalized. It validates request shape, preserves transaction context, and returns clear `customer` feedback for each outcome. It delegates decision logic to `payment authorization` while protecting the user experience boundary.

Owned Responsibility

functional responsibility for `checkout handling`; includes decision and orchestration flow to `Payment Authorization`

Key Collaborators

`Payment Authorization`

Linked Requirements

`REQ-PAY-001`, `REQ-PAY-003`, `REQ-PAY-006`

Payment Authorization (FU-PAYMENT-AUTHORIZATION)

Payment authorization orchestrates the final transaction decision path. It coordinates `fraud scoring`, external bank interactions, and escalation to `support review` when needed. The unit returns deterministic outcomes so downstream behavior remains consistent and testable.

Owned Responsibility

functional responsibility for `payment authorization`; includes decision and orchestration flow to `Bank Gateway Endpoint` and decision and orchestration flow to `Risk Scoring`

Key Collaborators

`Bank Gateway Endpoint`, `Risk Scoring`, `Support Review`

Linked Requirements

`REQ-PAY-001`, `REQ-PAY-004`, `REQ-PAY-006`

5.6.2. Fraud Evaluation (FG-FRAUD)

Risk scoring and fraud audit domain.

The `fraud evaluation` group exists to reduce unsafe approvals while preserving throughput for legitimate `payments`. It combines automated `risk scoring` with manual `support review` so uncertain transactions move through a controlled escalation path. The group ensures `risk` decisions are explainable, auditable, and operationally actionable.

Risk Scoring (FU-RISK-SCORING)

Risk scoring computes and classifies transaction `risk` before approval decisions are finalized. It provides a stable scoring contract to authorization and supports audit context for later analysis. The unit is focused on decision quality, consistency, and policy-driven classification behavior.

Owned Responsibility

functional responsibility for `risk scoring`; includes decision and orchestration flow to `PostgreSQL Client Driver`

Key Collaborators

`PostgreSQL Client Driver`

Linked Requirements

`REQ-PAY-002`, `REQ-PAY-005`

Support Review (FU-SUPPORT-REVIEW)

Support review handles manual decisions for escalated or ambiguous payment cases. It gives `support` operators context to approve, reject, or request additional verification in a controlled flow. The unit ensures manual intervention remains auditable, policy-constrained, and operationally reliable.

Owned Responsibility

functional responsibility for `support review`

Key Collaborators

none

Linked Requirements

`REQ-PAY-004`

5.6.3. Platform (FG-PLATFORM)

Cluster provisioning and GitOps operations domain.

The `platform` group provides the runtime and deployment foundation used by all product capabilities. It owns environment provisioning, namespace and cluster boundaries, and continuous reconciliation behavior. The group enables safe delivery and reliable operations without owning payment business logic.

Cluster Provisioning (FU-CLUSTER-PROVISIONING)

Cluster provisioning creates the runtime substrate used by all application workloads. It establishes cluster and namespace structure, baseline controls, and environment-level readiness. This unit is responsible for predictable, repeatable infrastructure foundations.

Owned Responsibility

functional responsibility for `cluster provisioning`

Key Collaborators

none

Linked Requirements

no explicit authored `requirement` trigger

GitOps Operations (FU-GITOPS-OPERATIONS)

GitOps operations continuously reconciles intended release state with actual runtime state. It governs rollout flow, drift correction, and operational delivery confidence across payment and fraud workloads. The unit provides safe and observable release behavior as an ongoing operational capability.

Owned Responsibility

functional responsibility for `gitops operations`; includes decision and orchestration flow to `Cluster Provisioning` and decision and orchestration flow to `Helm Runtime Platform`

Key Collaborators

`Cluster Provisioning`, `Helm Runtime Platform`

Linked Requirements

no explicit authored `requirement` trigger

6. Communication View

Show who communicates with whom across key request paths, including interface edges and dependency handoffs.

6.1. What This View Answers

- Who talks to whom and over which interfaces?
- Which edges are synchronous, asynchronous, event-driven, or callback-oriented?
- Which communication paths are critical to the main request flow?

6.2. Coverage Gaps

Coverage summary: 3/4 units have direct evidence coverage in this view.

- `Support Review` has no direct inferred runtime/deployment evidence yet.

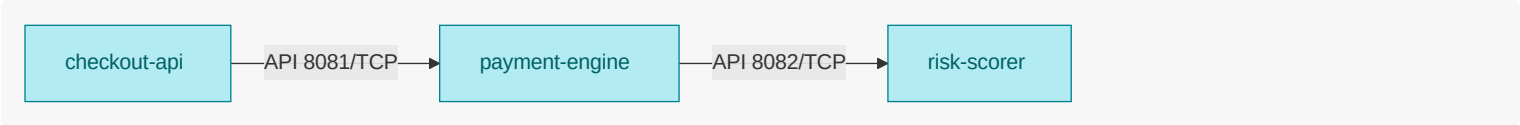
6.3. Recommended Next Evidence Additions

- Add explicit communication metadata (protocol, sync/async/event/callback) to `authored mappings` where currently implicit.
- Expose service protocol/port metadata in Helm or manifest values to enrich interface-edge inference.
- Capture missing communicating runtime units by adding discoverable workload artifacts under scanned inference roots.

6.4. Communication Path Graph

What this diagram shows: inferred unit-to-unit request/call flow, including available interface/port labels.

> **NOTE** This graph is inferred from available deployment/runtime artifacts and may be partial. Missing edges or interface details indicate missing derived communication evidence, not necessarily missing behavior.



6.5. Functional Groups and Units (View Scoped)

6.5.1. Payments (FG-PAYMENTS)

`payments` runs the public request path and keeps transaction context intact across runtime hops. The runtime design emphasizes low latency, deterministic request sequencing, and clear operational signals when dependencies slow down or fail.

Checkout Handling (FU-CHECKOUT)

`checkout handling` is the ingress runtime boundary and forwards normalized requests downstream. Runtime design emphasizes stable request contracts, context propagation, and graceful behavior under burst traffic and partial downstream degradation.

Inbound Interfaces

Interface	Description
`Customer`	Submits payment at `checkout`.

Outbound Interfaces

Interface	Description
`Payment Authorization`	Delegates `payment authorization`.

Communication Evidence

runtime: `payments`, `payments/checkout-api` | code modules: `checkout_api`

Payment Authorization (FU-PAYMENT-AUTHORIZATION)

`payment authorization` coordinates critical runtime dependencies and enforces decision-order determinism. Runtime design prioritizes idempotent transitions, explicit branch visibility, and controlled error normalization.

Inbound Interfaces

Interface	Description
`Checkout Handling`	Delegates `payment authorization`.

Outbound Interfaces

Interface	Description
`Bank Gateway Endpoint`	Calls external bank authorization endpoint.
`Risk Scoring`	Requests `fraud scoring` before final authorization.
`Support Review`	Escalates high-risk `payments` for `manual review`.

Communication Evidence

runtime: `payments/payment-engine` | code modules: domain, `payment_engine`

6.5.2. Fraud Evaluation (FG-FRAUD)

`fraud evaluation` runs as a critical dependency on the authorization path. Runtime behavior must return stable contracts quickly while preserving enough evidence for operator diagnosis and audit when high-risk classifications occur.

Risk Scoring (FU-RISK-SCORING)

`risk scoring` runs as a synchronous dependency with strict response-shape stability. Runtime design favors deterministic outputs, bounded latency, and clear degradation semantics for callers.

Inbound Interfaces

Interface	Description
`Payment Authorization`	Requests `fraud scoring` before final authorization.

Outbound Interfaces

Interface	Description
`PostgreSQL Client Driver`	Uses external library for audit persistence.

Communication Evidence

runtime: `risk`, `risk/risk-scorer` | code modules: `risk_scorer`, `support`

Support Review (FU-SUPPORT-REVIEW)

`support review` runtime optimizes for operator clarity and decision traceability over throughput. Runtime design emphasizes state visibility, reversible transitions where allowed, and reliable action logging.

Inbound Interfaces

Interface	Description
`Payment Authorization`	Escalates high-risk `payments` for `manual review`.
`Support Agent`	Performs manual payment reviews.

Outbound Interfaces

none

Communication Evidence

authored unit with no direct derived runtime/code evidence yet

7. Deployment View

Show inferred deployment artifacts and ownership mapping to authored units. This view emphasizes deployment relationships and `platform` operations.

7.1. What This View Answers

- How does authored intent become deployed runtime?
- What control-plane objects manage deployable units?
- Where are `platform` ownership boundaries and missing ownership signals?

7.2. Coverage Gaps

Coverage summary: 5/6 units have direct evidence coverage in this view.

- `Support Review` has no direct inferred runtime/deployment evidence yet.

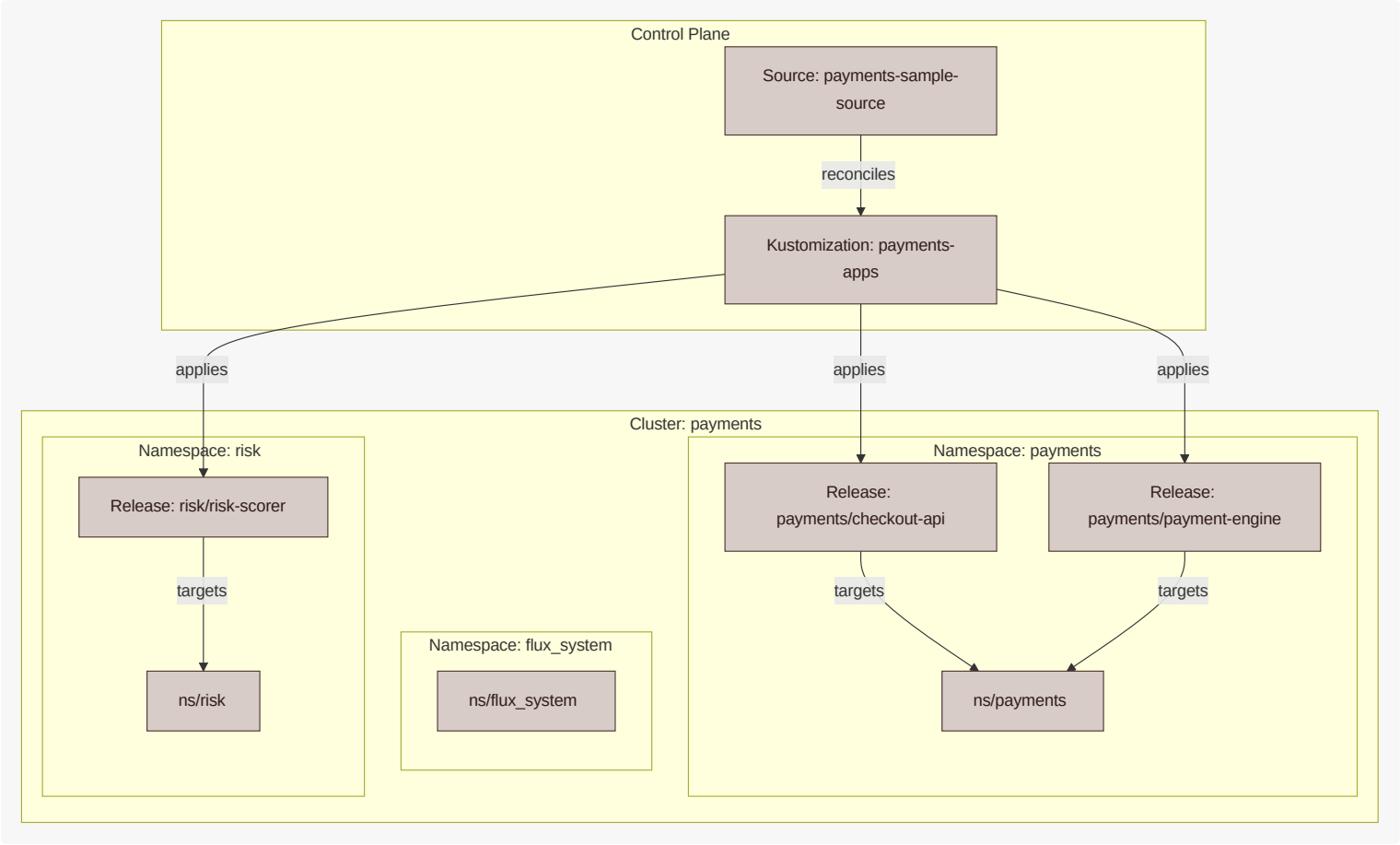
7.3. Recommended Next Evidence Additions

- Add owner metadata to Flux/Kustomization/Helm objects for clearer `platform` ownership mapping.
- Ensure deployment control artifacts (source, kustomization, release) are all included in scanned directories.
- Add deterministic naming conventions for release and namespace objects to improve cross-artifact linking.

7.4. Deployment Artifact Relationship Graph

What this diagram shows: how source/control-plane artifacts connect to releases, namespaces, workloads, and target cluster context.

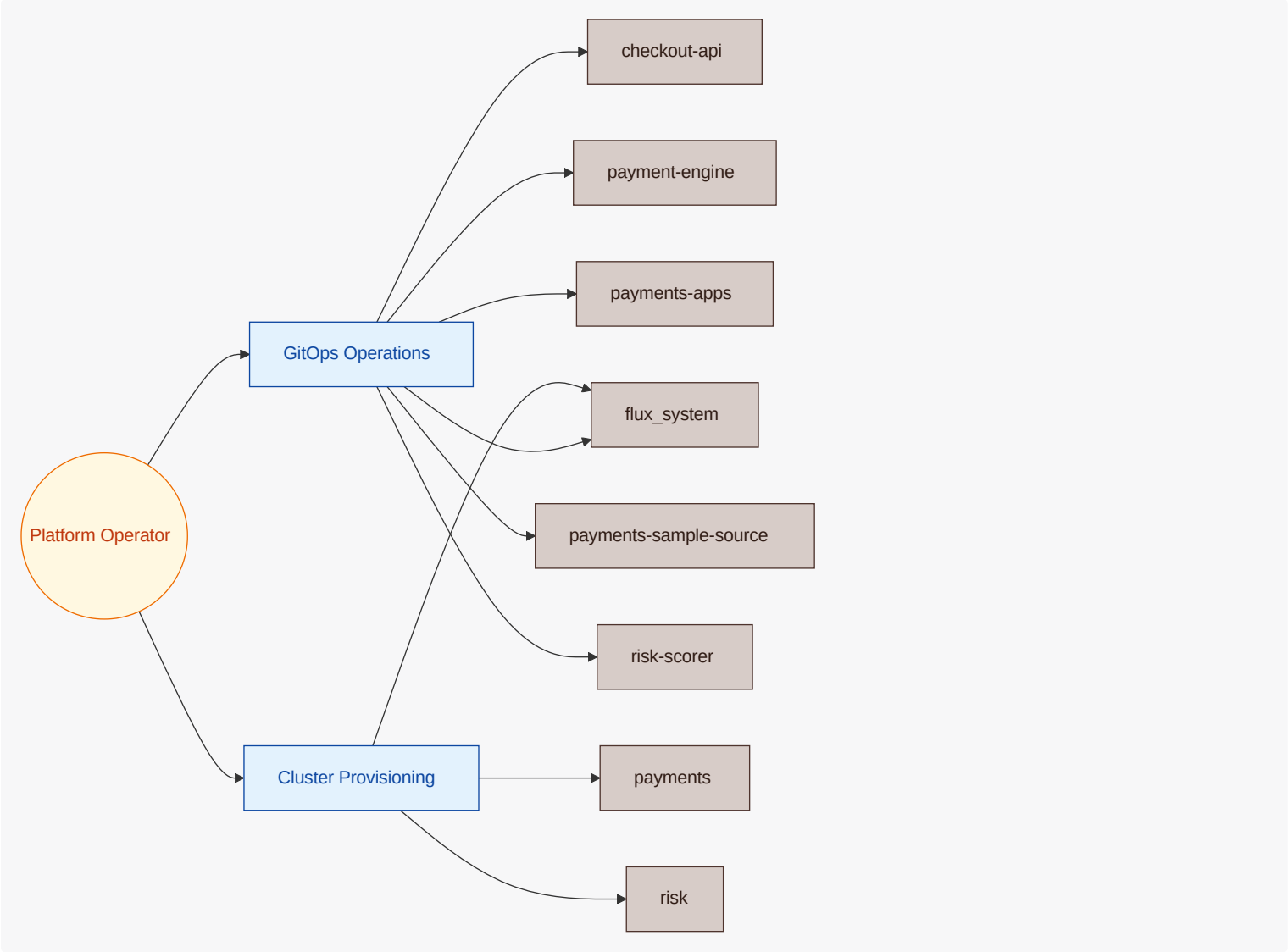
> **NOTE** This graph is inferred from available deployment artifacts (for example Terraform, Flux, Helm) and may be partial when source metadata is incomplete. Namespace and cluster boundaries are rendered as containers when target/placement relationships are inferred. Node labels include role hints (for example `Source`, `Kustomization`, `Release`, `Workload`, `ns/…`) to distinguish similarly named elements.



7.5. Platform Operations Graph

What this diagram shows: which platform actors operate which platform units and the control-plane targets they influence.

> **NOTE** Platform operations links are inferred from authored actor mappings plus discovered deployment control-plane artifacts.



7.6. Functional Groups and Units (View Scoped)

7.6.1. Payments (FG-PAYMENTS)

`payments` is deployed so `checkout` and authorization can roll independently with minimal blast radius. Deployment design prioritizes safe promotion, fast rollback, and dependency readiness checks before policy-impacting releases are exposed.

Checkout Handling (FU-CHECKOUT)

Checkout Handling Deployment Design

`checkout handling` is released independently to protect user-facing velocity without destabilizing policy logic. Deployment design favors fast rollback and compatibility checks with the current authorization interface.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-PAY-001`, `REQ-PAY-003`, `REQ-PAY-006`
Deployment dependencies	`Payment Authorization`
Inferred deployment evidence	runtime: `payments`, `payments/checkout-api`

Payment Authorization (FU-PAYMENT-AUTHORIZATION)

Payment Authorization Deployment Design

`payment authorization` deployment is policy-sensitive and therefore rollout-cautious. Releases should stage gradually, verify dependency health, and preserve in-flight transaction continuity.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-PAY-001`, `REQ-PAY-004`, `REQ-PAY-006`

Deployment Aspect	View-Scoped Detail
Deployment dependencies	`Bank Gateway Endpoint`, `Risk Scoring`, `Support Review`
Inferred deployment evidence	runtime: `payments/payment-engine`

7.6.2. Fraud Evaluation (FG-FRAUD)

`fraud evaluation` deployment is tuned for cautious rollout because scoring changes can alter approval rates. Deployment design favors staged promotions and rapid monitoring of scoring outcomes before wider exposure.

Risk Scoring (FU-RISK-SCORING)

Risk Scoring Deployment Design

`risk scoring` deployment uses staged rollout because scoring changes have immediate decision impact. Deployment design includes observation windows for score distribution and escalation-rate changes.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-PAY-002`, `REQ-PAY-005`
Deployment dependencies	`PostgreSQL Client Driver`
Inferred deployment evidence	runtime: `risk`, `risk/risk-scorer`

Support Review (FU-SUPPORT-REVIEW)

Support Review Deployment Design

`support review` deployment isolates privileged workflows from public ingress components. Deployment design must preserve active review sessions across upgrades and ensure queue/review latency visibility.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-PAY-004`
Deployment dependencies	none
Inferred deployment evidence	authored unit with no direct derived runtime/code evidence yet

7.6.3. Platform (FG-PLATFORM)

`platform` deployment design governs how infrastructure intent is applied and kept in sync over time. It prioritizes deterministic sequencing, drift control, and recoverable change paths before application teams depend on new `platform` states.

Cluster Provisioning (FU-CLUSTER-PROVISIONING)

Cluster Provisioning Deployment Design

`cluster provisioning` deployment is infrastructure-first and sequence-sensitive. Deployment design requires deterministic change plans, drift awareness, and rollback strategy before apply.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	no explicit authored `requirement` trigger
Deployment dependencies	none
Inferred deployment evidence	runtime: `payments`

GitOps Operations (FU-GITOPS-OPERATIONS)

GitOps Operations Deployment Design

`gitops operations` deployment defines the release control plane for application artifacts. Deployment design prioritizes policy gates, controlled promotion, and resilient rollback mechanics.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	no explicit authored `requirement` trigger
Deployment dependencies	`Cluster Provisioning`, `Helm Runtime Platform`
Inferred deployment evidence	runtime: `flux-system/payments-apps`, `flux-system/payments-sample-source`, `flux_system`

8. Security View

Show inferred exposure and dependency `risk` points aligned to unit boundaries, focused on attack paths and security evidence.

8.1. What This View Answers

- What are the main trust boundaries and attack paths?
- Which units are exposed and what security evidence exists?
- Where is the threat model incomplete?

8.2. Coverage Gaps

Coverage summary: 4/6 units have direct evidence coverage in this view.

- `Cluster Provisioning` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- `GitOps Operations` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.

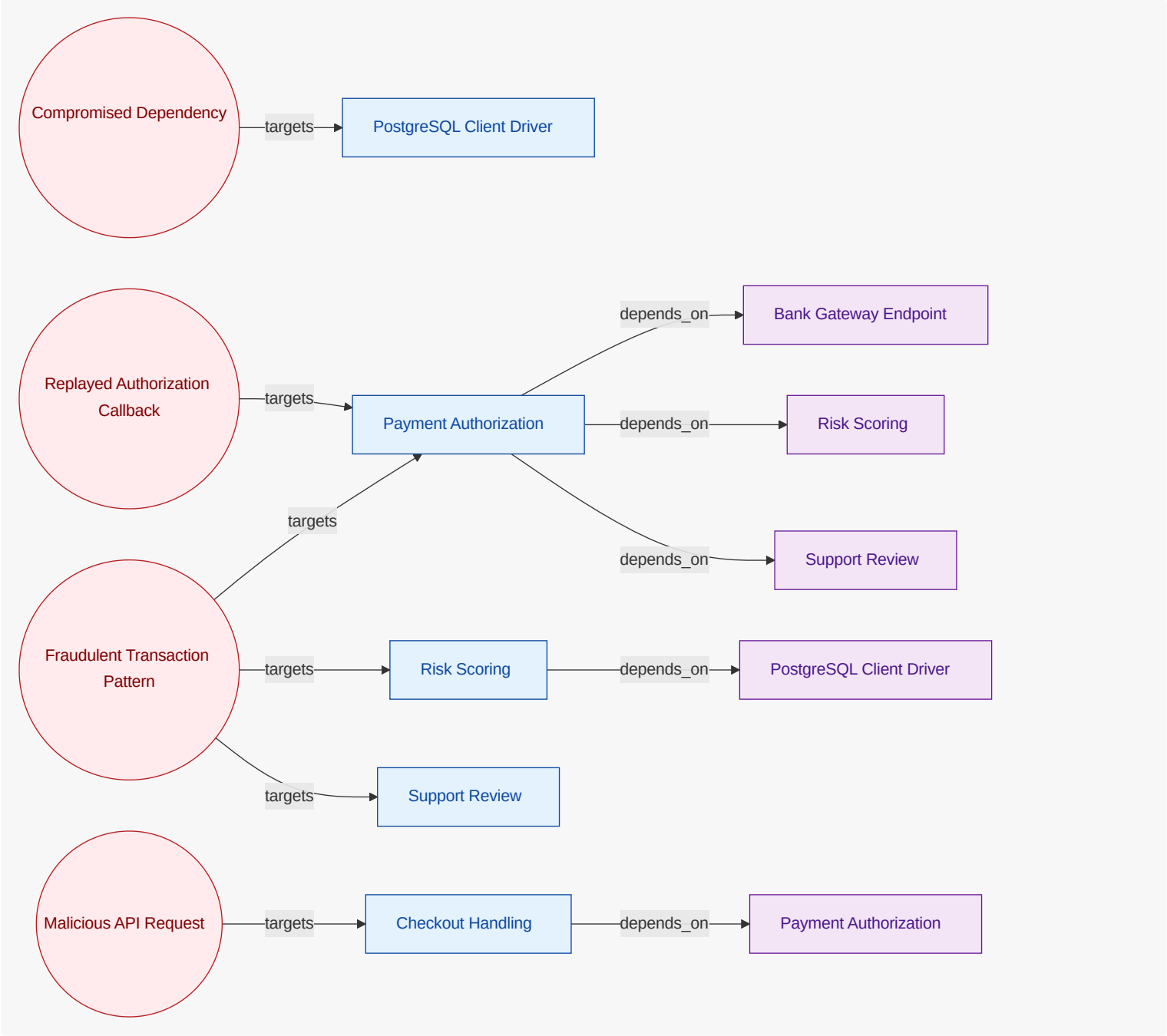
8.3. Recommended Next Evidence Additions

- Add explicit authored attack-vector mappings for units currently marked without threat linkage.
- Add security signal ownership metadata (logs/alerts/`events`) for unresolved observability evidence.
- Map additional security-relevant dependencies and exposures into `authored mappings` for traceable coverage.

8.4. Attack Path Diagram

What this diagram shows: authored attack vectors, their targeted units, and security-relevant dependency paths.

> **NOTE**
> Security relationships shown here are explicitly modeled or inferred from available evidence and may be incomplete. Absence of a path does not imply absence of risk. Threats that are currently mitigated (including fraud/abuse threats) should still be modeled as attack vectors for auditability and control-trace visibility.



8.5. Security Logging and Observability

Signal	Layer	Owner	Evidence
application security telemetry hooks	code	FU-RISK-SCORING	code support/audit_envelope.ts
fraud and abuse detection code paths	code	FU-RISK-SCORING	code risk_scorer.ts
deployment and platform audit events	deployment	FU-CHECKOUT	helmrelease checkout-api
deployment and platform audit events	deployment	FU-CHECKOUT	namespace payments
deployment and platform audit events	deployment	FU-CLUSTER-PROVISIONING	cluster payments
deployment and platform audit events	deployment	FU-GITOPS-OPERATIONS	gitrepository payments-sample-source
deployment and platform audit events	deployment	FU-GITOPS-OPERATIONS	kustomization payments-apps

Signal	Layer	Owner	Evidence
deployment and platform audit events	deployment	FU-GITOPS-OPERATIONS	namespace flux_system
deployment and platform audit events	deployment	FU-PAYMENT-AUTHORIZATION	helmrelease payment-engine
deployment and platform audit events	deployment	FU-RISK-SCORING	helmrelease risk-scorer
deployment and platform audit events	deployment	FU-RISK-SCORING	namespace risk
abuse and fraud decision logs	runtime	FU-RISK-SCORING	helmrelease risk-scorer
abuse and fraud decision logs	runtime	FU-RISK-SCORING	namespace risk

8.6. Functional Groups and Units (View Scoped)

8.6.1. Payments (FG-PAYMENTS)

`payments` protects public trust boundaries where untrusted input and external callbacks enter the system. Security design focuses on strict validation, replay resistance, and safe failure behavior that preserves `customer` trust during degraded dependency conditions.

Checkout Handling (FU-CHECKOUT)

`checkout handling` security protects the internet-facing ingress. The design requires strict validation, abuse resistance, and traceable rejection behavior so malicious inputs are contained without harming legitimate `customer` flows.

Attack Surface

`Payment Authorization`

Main Threats

`Malicious API Request`

Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

Security Evidence Present

runtime: `payments`, `payments/checkout-api` | code modules: `checkout_api`

Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

Payment Authorization (FU-PAYMENT-AUTHORIZATION)

`payment authorization` security protects decision integrity and callback integrity. The design defends against replay, policy probing, and ambiguous error channels that could be abused.

Attack Surface

`Bank Gateway Endpoint`, `Risk Scoring`, `Support Review`

Main Threats

`Fraudulent Transaction Pattern`, `Replayed Authorization Callback`

Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

Security Evidence Present

runtime: `payments/payment-engine` | code modules: domain, `payment_engine`

Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

8.6.2. Fraud Evaluation (FG-FRAUD)

`fraud evaluation` security focuses on decision integrity and review integrity. The design assumes attempted manipulation of `risk` inputs, dependency trust, and `manual review` workflows, and therefore requires strong evidence lineage and tamper-evident operator actions.

Risk Scoring (FU-RISK-SCORING)

`risk scoring` security assumes adversarial input attempts and model manipulation pressure. Design controls enforce strict input handling and evidence capture for post-event investigation.

Attack Surface

`PostgreSQL Client Driver`

Main Threats

`Fraudulent Transaction Pattern`

Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

Security Evidence Present

runtime: `risk`, `risk/risk-scorer` | code modules: `risk_scorer`, `support`

Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

Support Review (FU-SUPPORT-REVIEW)

`support review` security requires strong operator identity controls and tamper-evident audit trails. The design prevents silent overrides and guarantees every manual decision is attributable.

Attack Surface

none

Main Threats

`Fraudulent Transaction Pattern`

Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

Security Evidence Present

authored unit with no direct derived runtime/code evidence yet

Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

8.6.3. Platform (FG-PLATFORM)

`platform` security design establishes baseline controls inherited by all higher-layer units. It focuses on trusted reconciliation, isolation defaults, and policy enforcement points where compromise would have broad systemic impact.

Cluster Provisioning (FU-CLUSTER-PROVISIONING)

`cluster provisioning` security sets inherited baseline hardening and isolation defaults. Design must detect and remediate security drift early because downstream controls depend on this layer.

Attack Surface

none

Main Threats

no explicit `attack vector` targeting this unit

Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

Security Evidence Present

runtime: `payments`

Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

GitOps Operations (FU-GITOPS-OPERATIONS)

`gitops operations` security protects source trust and reconciliation authority. The design prevents unauthorized state changes and enforces policy before runtime apply occurs.

Attack Surface

`Cluster Provisioning`, `Helm Runtime Platform`

Main Threats

no explicit `attack vector` targeting this unit

Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

Security Evidence Present

runtime: `flux-system/payments-apps`, `flux-system/payments-sample-source`, `flux_system`

Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

9. Traceability View

Show requirement-to-unit-to-evidence traceability, including coverage confidence and explicit evidence gaps.

9.1. What This View Answers

- Which `requirements` map to which units?
- Which authored units have runtime/code evidence and which do not?
- Where are confidence or completeness gaps in trace coverage?

9.2. Coverage Gaps

Coverage summary: 5/6 units have direct evidence coverage in this view.

- `Cluster Provisioning` has no direct inferred code evidence yet.
- `GitOps Operations` has no direct inferred code evidence yet.
- `Support Review` has no direct inferred code evidence yet.
- `Support Review` has no direct inferred runtime evidence yet.

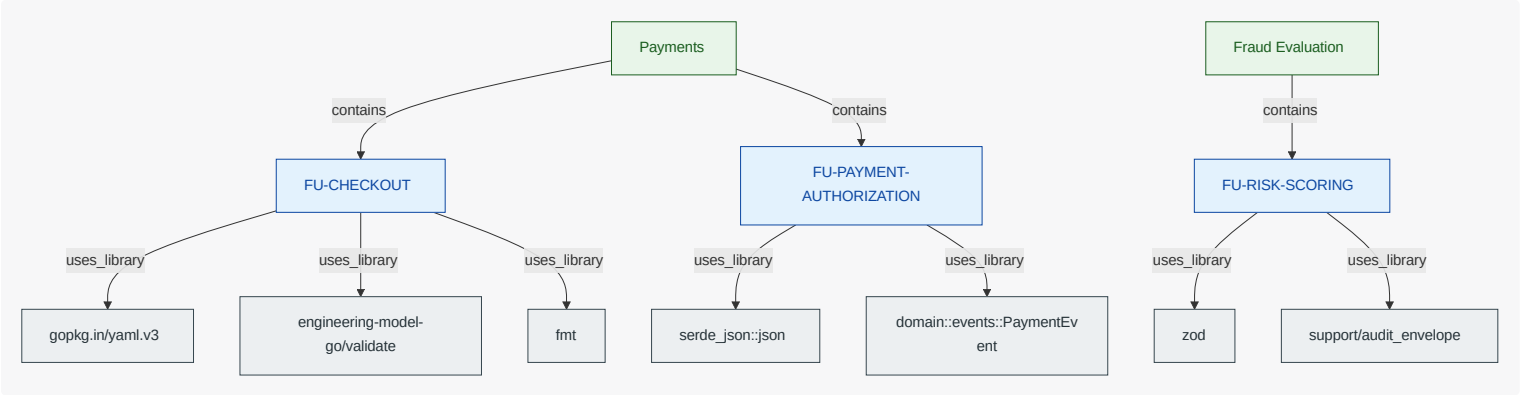
9.3. Recommended Next Evidence Additions

- Add explicit requirement-to-owner mappings where `requirement` scope is currently broad or ambiguous.
- Add ownership annotations for runtime and code artifacts where coverage is still unresolved.
- Prioritize closing gaps for units that are authored but lack both runtime and code evidence.

9.4. Traceability Evidence Graph

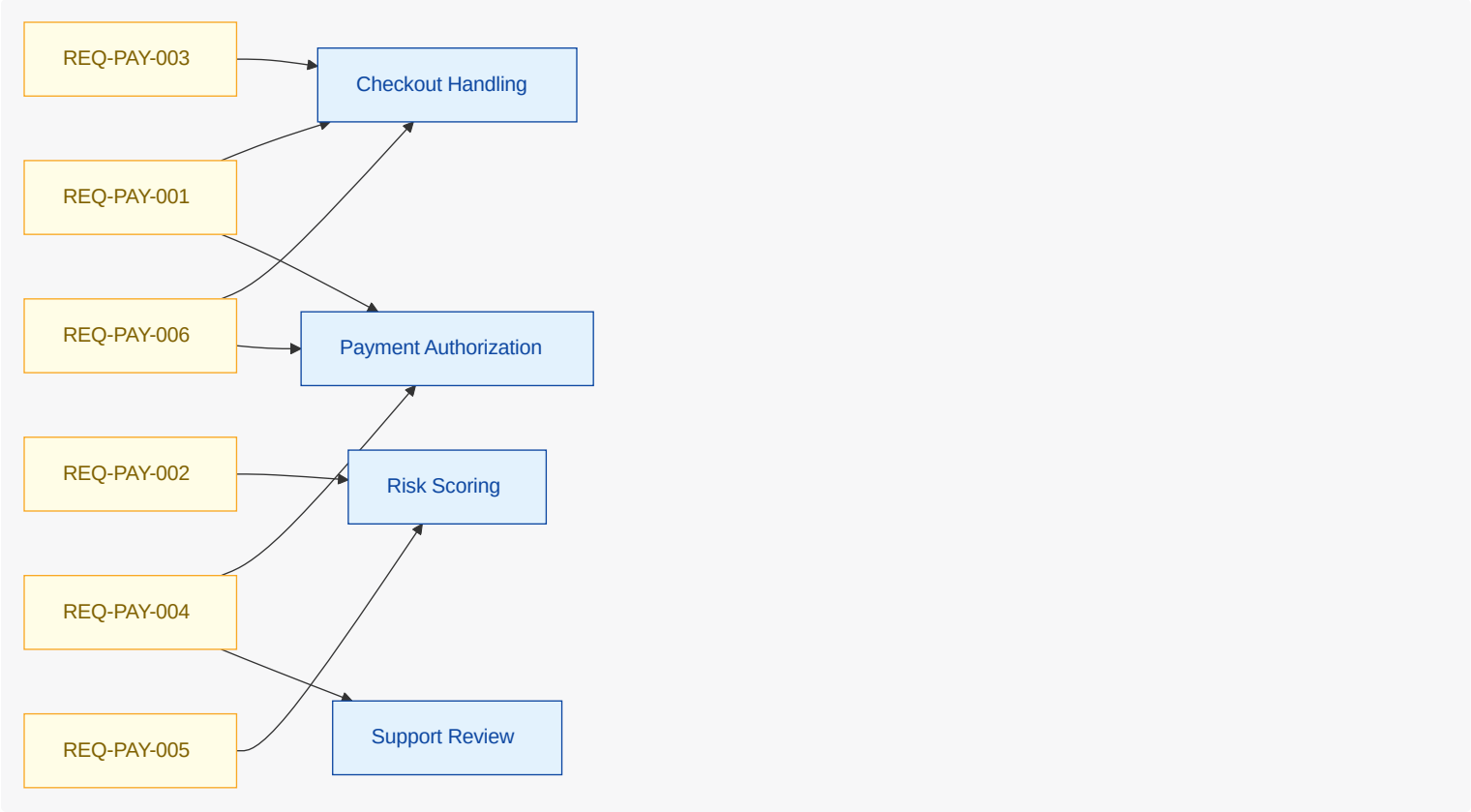
What this diagram shows: authored units and the inferred evidence links that support traceability confidence.

> **NOTE** This graph summarizes inferred ownership/evidence links and is intended for traceability confidence checks.



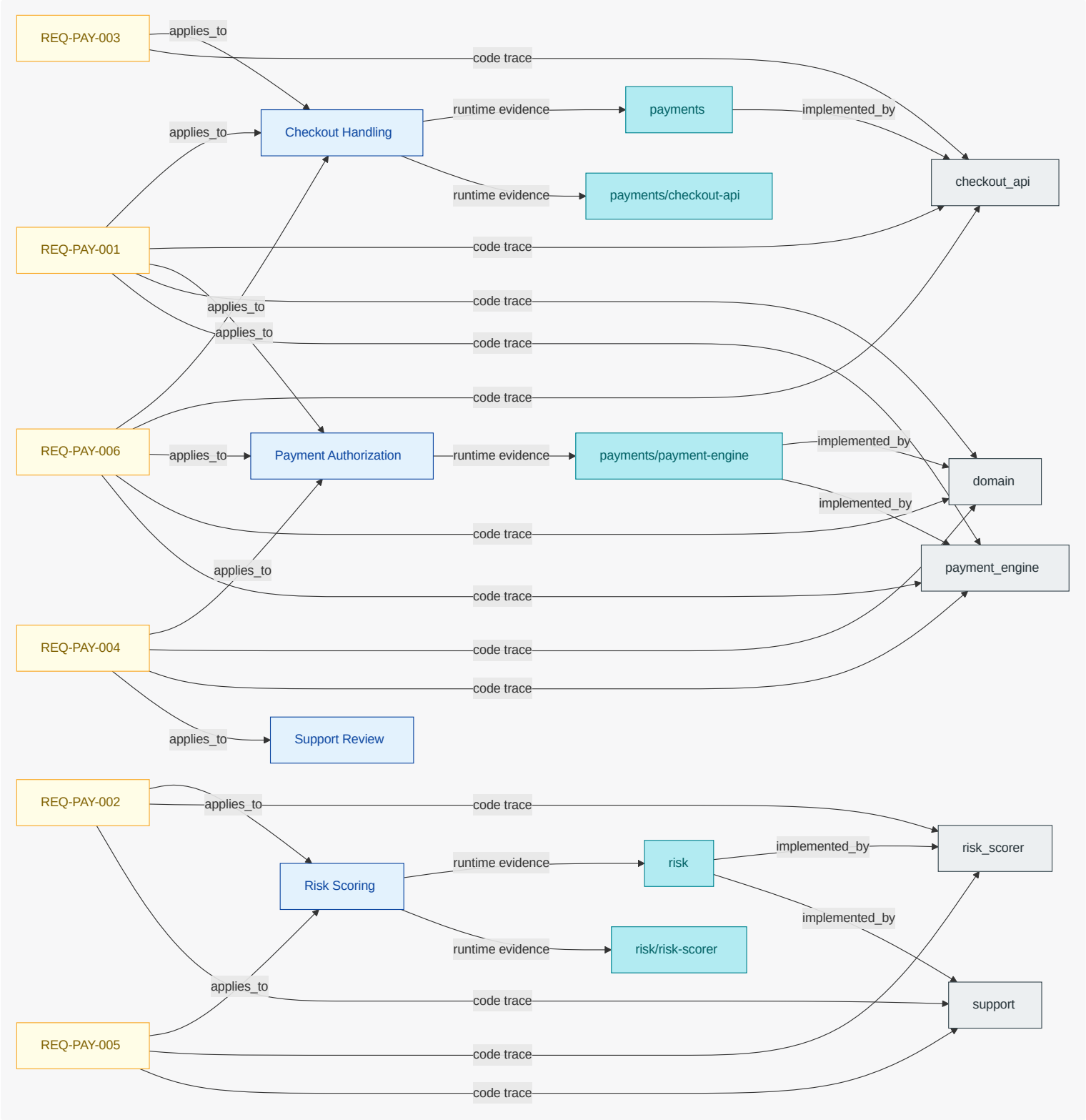
9.5. Requirement-to-Unit Mapping

What this diagram shows: direct authored mapping from requirements to the functional units they apply to.



9.6. Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: requirement-to-unit links extended with inferred runtime/code evidence attached to each unit.



Show requirement-to-unit mappings inferred from appliesTo and authored architecture ownership boundaries.

10. Traceability Appendix

10.1. Requirement Details

10.1.1. REQ-PAY-001

While `checkout is active`, when `payment authorization is requested`, the `payments system` shall create a payment session and request authorization from bank gateway. Notes: Core authorization path from `checkout` to bank gateway.

10.1.2. REQ-PAY-002

Where `fraud check is enabled`, when `payment authorization is requested`, the `fraud detection system` shall calculate a `risk score` before approval. Notes: Fraud scoring gate before approval.

10.1.3. REQ-PAY-003

If `payment is declined`, then the `payments system` shall present a `decline reason` and a retry option to `customer`. Notes: Decline handling and `customer` feedback.

10.1.4. REQ-PAY-004

While `checkout is active`, when `payment authorization is requested` and `risk score is high`, the `payments system` shall place the payment in `review pending` state and notify `support agent`. Notes: Escalation flow for high-risk `payments`.

10.1.5. REQ-PAY-005

Where `fraud check is enabled`, when `payment authorization is requested`, the `payments system` shall persist an audit record with `payment id` and `risk score`. Notes: Auditability for authorization decisions.

10.1.6. REQ-PAY-006

If `bank link is unavailable`, then the `payments system` shall present a temporary unavailable message to `customer`. Notes: Fallback behavior when external banking path is down.

10.2. Verification Inventory

What this section shows: verification checks inferred from test artifacts, linked to test code elements and mapped to verified requirements.

10.2.1. Authorized Checkout Flow

Field	Value
ID	`VER-INF-E2E-AUTHORIZED_CHECKOUT_FLOW-8C6597`
Kind	e2e
Status	not-run
Verifies Requirements	`REQ-PAY-001`
Test Code Elements	tests/e2e/authorized_checkout_flow.yaml
Derived Functional Owners	`FU-CHECKOUT`, `FU-PAYMENT-AUTHORIZATION`
Result Summary	not-run:1
Evidence	tests/e2e/authorized_checkout_flow.yaml
Notes	executes authorized `checkout` flow from session start through `payment authorization` success

10.2.2. Audit Record Persistence Test

Field	Value
ID	`VER-INF-INTEGRATION-AUDIT_RECORD_PERSISTENCE_TEST-B333A2`
Kind	integration
Status	partial
Verifies Requirements	`REQ-PAY-005`
Test Code Elements	tests/integration/audit_record_persistence_test.ts
Derived Functional Owners	`FU-RISK-SCORING`
Result Summary	partial:1
Evidence	test-results/audit-persistence-report.md, tests/integration/audit_record_persistence_test.ts
Notes	checks audit record persistence includes `payment id` and computed `risk score`

10.2.3. Fraud Gate Test

Field	Value
ID	`VER-INF-INTEGRATION-FRAUD_GATE_TEST-12F3BD`
Kind	integration
Status	pass
Verifies Requirements	`REQ-PAY-002`, `REQ-PAY-004`
Test Code Elements	tests/integration/fraud_gate_test.rs
Derived Functional Owners	`FU-PAYMENT-AUTHORIZATION`, `FU-RISK-SCORING`, `FU-SUPPORT-REVIEW`
Result Summary	pass:2
Evidence	test-results/integration-fraud-gate.json, tests/integration/fraud_gate_test.rs
Notes	validates fraud gate escalation and block decisions before authorization completion

10.2.4. Contract Checkout Decline

Field	Value
ID	`VER-INF-TEST-CONTRACT_CHECKOUT_DECLINE-5D1F26`
Kind	test
Status	pass
Verifies Requirements	`REQ-PAY-003`, `REQ-PAY-006`
Test Code Elements	none
Derived Functional Owners	`FU-CHECKOUT`, `FU-PAYMENT-AUTHORIZATION`
Result Summary	pass:2
Evidence	test-results/contract-checkout-decline.json
Notes	Inferred from test result artifact.

10.3. Verification Result Mapping

What this section shows: per-requirement result mapping inferred from test-result artifacts.

Check ID	Requirement	Result	Evidence / Notes
`VER-INF-E2E-AUTHORIZED_CHECKOUT_FLOW-8C6597`	`REQ-PAY-001`	not-run	none No parsed test-result artifact for this `requirement` yet.
`VER-INF-INTEGRATION-FRAUD_GATE_TEST-12F3BD`	`REQ-PAY-002`	pass	test-results/integration-fraud-gate.json
`VER-INF-TEST-CONTRACT_CHECKOUT_DECLINE-5D1F26`	`REQ-PAY-003`	pass	test-results/contract-checkout-decline.json
`VER-INF-INTEGRATION-FRAUD_GATE_TEST-12F3BD`	`REQ-PAY-004`	pass	test-results/integration-fraud-gate.json
`VER-INF-INTEGRATION-AUDIT_RECORD_PERSISTENCE_TEST-B333A2`	`REQ-PAY-005`	partial	test-results/audit-persistence-report.md
`VER-INF-TEST-CONTRACT_CHECKOUT_DECLINE-5D1F26`	`REQ-PAY-006`	pass	test-results/contract-checkout-decline.json

11. Reference Index

11.1. Authored References

ACT-CUSTOMER

Kind: Actor Name: Customer

Starts checkout and confirms payment.

ACT-PLATFORM-OPERATOR

Kind: Actor Name: Platform Operator

Operates GitOps and platform lifecycle.

ACT-SUPPORT

Kind: Actor Name: Support Agent

Reviews high-risk and declined-payment cases.

AV-COMPROMISED-DEPENDENCY

Kind: Attack Vector Name: Compromised Dependency

Supply chain compromise in external SDK or image.

AV-FRAUDULENT-TRANSACTION-PATTERN

Kind: Attack Vector Name: Fraudulent Transaction Pattern

Coordinated abuse pattern using synthetic identities, stolen instruments, and velocity anomalies to seek unauthorized approvals.

AV-MALICIOUS-API-REQUEST

Kind: Attack Vector Name: Malicious API Request

Crafted payloads attempting to bypass authorization controls.

AV-REPLAYED-AUTH-CALLBACK

Kind: Attack Vector Name: Replayed Authorization Callback

Replay attack against external authorization callback flow.

FG-FRAUD

Kind: Functional Group Name: Fraud Evaluation

Risk scoring and fraud audit domain.

FG-PAYMENTS

Kind: Functional Group Name: Payments

Core payment checkout and authorization domain.

FG-PLATFORM

Kind: Functional Group Name: Platform

Cluster provisioning and GitOps operations domain.

FU-CHECKOUT

Kind: Functional Unit Name: Checkout Handling

Checkout handling is the user-facing entrypoint where payment requests are initiated and normalized. It validates request shape, preserves transaction context, and returns clear customer feedback for each outcome. It delegates decision logic to payment authorization while protecting the user experience boundary.

FU-CLUSTER-PROVISIONING

Kind: Functional Unit Name: Cluster Provisioning

Cluster provisioning creates the runtime substrate used by all application workloads. It establishes cluster and namespace structure, baseline controls, and environment-level readiness. This unit is responsible for predictable, repeatable infrastructure foundations.

FU-GITOPS-OPERATIONS

Kind: Functional Unit Name: GitOps Operations

GitOps operations continuously reconciles intended release state with actual runtime state. It governs rollout flow, drift correction, and operational delivery confidence across payment and fraud workloads. The unit provides safe and observable release behavior as an ongoing operational capability.

FU-PAYMENT-AUTHORIZATION

Kind: Functional Unit Name: Payment Authorization

Payment authorization orchestrates the final transaction decision path. It coordinates fraud scoring, external bank interactions, and escalation to support review when needed. The unit returns deterministic outcomes so downstream behavior remains consistent and testable.

FU-RISK-SCORING

Kind: Functional Unit Name: Risk Scoring

Risk scoring computes and classifies transaction risk before approval decisions are finalized. It provides a stable scoring contract to authorization and supports audit context for later analysis. The unit is focused on decision quality, consistency, and policy-driven classification behavior.

FU-SUPPORT-REVIEW

Kind: Functional Unit Name: Support Review

Support review handles manual decisions for escalated or ambiguous payment cases. It gives support operators context to approve, reject, or request additional verification in a controlled flow. The unit ensures manual intervention remains auditable, policy-constrained, and operationally reliable.

REF-BANK-GATEWAY-ENDPOINT

Kind: Referenced Element Name: Bank Gateway Endpoint

external_service_endpoint / runtime

REF-HELM-PLATFORM

Kind: Referenced Element Name: Helm Runtime Platform

platform_service / runtime

REF-POSTGRES-DRIVER

Kind: Referenced Element Name: PostgreSQL Client Driver

third_party_library / code

REQ-PAY-001

Kind: Requirement Name: REQ-PAY-001 Resolves to: `canonical` While checkout is active, when payment authorization is requested, the payments system shall create a payment session and request authorization from bank gateway.

REQ-PAY-002

Kind: Requirement Name: REQ-PAY-002 Resolves to: `canonical` Where fraud check is enabled, when payment authorization is requested, the fraud detection system shall calculate a risk score before approval.

REQ-PAY-003

Kind: Requirement Name: REQ-PAY-003 Resolves to: `canonical` If payment is declined, then the payments system shall present a decline reason and a retry option to customer.

REQ-PAY-004

Kind: Requirement Name: REQ-PAY-004 Resolves to: `canonical` While checkout is active, when payment authorization is requested and risk score is high, the payments system shall place the payment in review pending state and notify support agent.

REQ-PAY-005

Kind: Requirement Name: REQ-PAY-005 Resolves to: `canonical` Where fraud check is enabled, when payment authorization is requested, the payments system shall persist an audit record with payment id and risk score.

REQ-PAY-006

Kind: Requirement Name: REQ-PAY-006 Resolves to: `canonical` If bank link is unavailable, then the payments system shall present a temporary unavailable message to customer.

11.2. Catalog References

customer

End user submitting payment details during checkout. Canonical: `open`

platform operator

Role responsible for GitOps and platform lifecycle operations. Canonical: `open`

support agent

Operations role that investigates declines and high-risk review cases. Canonical: `open`

3DS verification is required

Condition indicating strong customer authentication must be performed. Canonical: `open`

payment authorization

Coordinates authorization flow and bank gateway interaction. Canonical: `open`

bank gateway endpoint

External banking endpoint used for card authorization and decline decisions. Canonical: `open`

bank link is unavailable

Event raised when payment engine cannot reach the external bank gateway. Canonical: `open`

bank link is available

Connectivity state showing bank gateway communication is currently healthy. Canonical: `open`

customer

End user submitting payment details during checkout. Canonical: `open`

replayed authorization callback

Replay of a valid callback to induce duplicate or unauthorized state transitions. Canonical: `open`

checkout handling

Handles checkout session start and customer-facing payment messaging. Canonical: `open`

checkout is active

Processing state where the user is actively submitting or confirming payment. Canonical: `open`

cluster provisioning

Provides cluster-level platform setup and environment bootstrap. Canonical: `open`

gitops operations

Operates declarative delivery and release management flows. Canonical: `open`

decline reason

Structured reason code or message explaining why authorization failed. Canonical: `open`

compromised dependency

Supply-chain compromise in runtime image or code dependency. Canonical: `open`

fraudulent transaction pattern

Coordinated abuse pattern using synthetic identities, stolen instruments, or velocity anomalies to obtain unauthorized approvals. Canonical: `open`

fraud check is enabled

Feature flag controlling whether risk scoring is required before authorization. Canonical: `open`

fraud evaluation

Functional group for risk scoring and fraud review capabilities. Canonical: `open`

fraud evaluation

Functional group for risk scoring and fraud review capabilities. Canonical: `open`

risk scoring

Evaluates transaction risk and returns score classification. Canonical: `open`

helm platform

Runtime packaging and deployment substrate for workloads. Canonical: `open`

risk score is high

Event emitted when computed fraud risk exceeds the configured high-risk threshold. Canonical: `open`

malicious api request

Crafted payload attempting to bypass validation or authorization logic. Canonical: `open`

support review

Supports manual review and escalation handling for risky transactions. Canonical: `open`

normal mode is active

Standard operating mode with full external dependency usage enabled. Canonical: `open`

payment authorization is requested

Event raised when checkout asks payment engine to authorize a transaction. Canonical: `open`

payment is declined

Event indicating authorization was rejected and customer decline handling must execute. Canonical: `open`

fraudulent transaction pattern

Coordinated abuse pattern using synthetic identities, stolen instruments, or velocity anomalies to obtain unauthorized approvals. Canonical: `open`

payments

Functional group for checkout and authorization capabilities. Canonical: `open`

payments

Functional group for checkout and authorization capabilities. Canonical: `open`

platform

Functional group for provisioning and platform operations capabilities. Canonical: `open`

platform operator

Role responsible for GitOps and platform lifecycle operations. Canonical: `open`

postgresql client driver

Third-party library used for application data persistence connectivity. Canonical: `open`

payment review is pending

Transaction state requiring manual support review before final disposition. Canonical: `open`

risk threshold is exceeded

Condition indicating risk score crossed a policy threshold for escalation. Canonical: `open`

support agent

Operations role that investigates declines and high-risk review cases. Canonical: `open`

support agent is notified

Event indicating manual review notification was sent to support operations. Canonical: `open`

compromised dependency

Supply-chain compromise in runtime image or code dependency. Canonical: `open`

fraudulent transaction pattern

Coordinated abuse pattern using synthetic identities, stolen instruments, or velocity anomalies to obtain unauthorized approvals. Canonical: `open`

malicious api request

Crafted payload attempting to bypass validation or authorization logic. Canonical: `open`

replayed authorization callback

Replay of a valid callback to induce duplicate or unauthorized state transitions. Canonical: `open`

3DS verification is required

Condition indicating strong customer authentication must be performed. Canonical: `open`

risk threshold is exceeded

Condition indicating risk score crossed a policy threshold for escalation. Canonical: `open`

decline reason

Structured reason code or message explaining why authorization failed. Canonical: `open`

payment id

Stable identifier used to correlate a payment across services and records. Canonical: `open`

risk score

Numeric fraud score produced by risk evaluation logic. Canonical: `open`

bank link is unavailable

Event raised when payment engine cannot reach the external bank gateway. Canonical: `open`

payment authorization is requested

Event raised when checkout asks payment engine to authorize a transaction. Canonical: `open`

payment is declined

Event indicating authorization was rejected and customer decline handling must execute. Canonical: `open`

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Event emitted when computed fraud risk exceeds the configured high-risk threshold. Canonical: `open`

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Runtime packaging and deployment substrate for workloads. Canonical: `open`

postgresql client driver

Third-party library used for application data persistence connectivity. Canonical: `open`

bank link is available

Connectivity state showing bank gateway communication is currently healthy. Canonical: `open`

checkout is active

Processing state where the user is actively submitting or confirming payment. Canonical: `open`

payment review is pending

Transaction state requiring manual support review before final disposition. Canonical: `open`

actor

A person or operational role that interacts with functional units. Canonical: `open`

attack vector

A technical misuse or attack path that targets functional, referenced, or runtime elements. Canonical: `open`

authored mapping

An explicit relationship declared in architecture inputs between authored or referenced elements. Canonical: `open`

code element

An inferred code structure or ownership element discovered from source trees and build metadata. Canonical: `open`

functional group

A major authored capability area that groups related functional units. Canonical: `open`

functional unit

An authored working unit inside a functional group that owns specific behavior. Canonical: `open`

inferred mapping

A discovered relationship that links inferred runtime/code elements upward to authored design. Canonical: `open`

referenced element

An architecture-relevant external, platform, or third-party dependency represented by role. Canonical: `open`

requirement

A structured requirement statement that defines expected system behavior and maps to functional ownership. Canonical: `open`

runtime element

An inferred runtime realization element discovered from infrastructure and deployment sources. Canonical: `open`

upward linking

Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs. Canonical: `open`

verification check

An inferred or authored verification artifact that validates requirement behavior with test evidence. Canonical: `open`

11.3. Inferred Runtime References

ID	Kind	Owner	Description	Source
payments	Runtime cluster	FU-CLUSTER-PROVISIONING	provisions the shared EKS control plane that hosts payments and risk workloads	examples/payments-engineering-sample/infra/terraform/main.tf
flux-system/payments-sample-source	Runtime gitrepository	FU-GITOPS-OPERATIONS	tracks the desired repository state that drives Flux reconciliation for payments workloads	examples/payments-engineering-sample/infra/flux/sources/payments-source.yaml
payments/checkout-api	Runtime helmrelease	FU-CHECKOUT	exposes checkout ingress and forwards normalized payment requests to authorization flow	examples/payments-engineering-sample/infra/flux/releases/checkout-api.yaml
payments/payment-engine	Runtime helmrelease	FU-PAYMENT-AUTHORIZATION	executes payment authorization decisions and coordinates fraud checks plus bank callbacks	examples/payments-engineering-sample/infra/flux/releases/payment-engine.yaml
risk/risk-scorer	Runtime helmrelease	FU-RISK-SCORING	computes fraud risk score signals and persists evidence used by authorization decisions	examples/payments-engineering-sample/infra/flux/releases/risk-scorer.yaml
flux-system/payments-apps	Runtime kustomization	FU-GITOPS-OPERATIONS	applies and reconciles HelmRelease definitions for checkout, payment, and risk services	examples/payments-engineering-sample/infra/flux/kustomizations/payments-apps.yaml
flux_system	Runtime namespace	FU-GITOPS-OPERATIONS	control-plane namespace where Flux GitOps operators reconcile deployment state	examples/payments-engineering-sample/infra/terraform/main.tf
payments	Runtime namespace	FU-CHECKOUT	runtime namespace boundary for checkout and payment authorization workloads	examples/payments-engineering-sample/infra/terraform/main.tf
risk	Runtime namespace	FU-RISK-SCORING	runtime namespace boundary for fraud scoring and review support workloads	examples/payments-engineering-sample/infra/terraform/main.tf

11.4. Inferred Code References

ID	Kind	Owner	Description	Source
gopkg.in/yaml.v3	Code library_external	FU-CHECKOUT	n/a	checkout_
serde_json::json	Code library_external	FU-PAYMENT-AUTHORIZATION	n/a	payment_
zod	Code library_external	FU-RISK-SCORING	n/a	risk_score
./support/audit_envelope	Code library_first_party	FU-RISK-SCORING	n/a	risk_score
[[REF_CODE_LIBRARY_FIRST_PARTY_CRATE*DOMAIN*EVENTS__PAYMENTEVENT]]crate::domain::events::PaymentEvent	Code library_first_party	FU-PAYMENT-AUTHORIZATION	n/a	payment_
github.com/labeth/engineering-model-go/validate	Code library_first_party	FU-CHECKOUT	n/a	checkout_

ID	Kind	Owner	Description	Source
fmt	Code library_stdlib	FU-CHECKOUT	n/a	checkout_
checkout_api.go	Code source_file	FU-CHECKOUT	accepts checkout requests and forwards payment authorization flow input	checkout_
domain/events.rs	Code source_file	FU-PAYMENT-AUTHORIZATION	defines payment domain events used across authorization and risk paths	domain/e
payment_engine.rs	Code source_file	FU-PAYMENT-AUTHORIZATION	executes authorization decisions and persists payment-side audit records	payment_
risk_scorer.ts	Code source_file	FU-RISK-SCORING	computes fraud risk scores and decides review or decline outcomes	risk_score
support/audit_envelope.ts	Code source_file	FU-RISK-SCORING	formats support-review audit envelopes for traceability workflows	support/a
CODE-AUTHORIZEPAYMENT	Code symbol	FU-PAYMENT-AUTHORIZATION	n/a	payment_
CODE-BUILDREVIEWREASON	Code symbol	FU-RISK-SCORING	n/a	risk_score
CODE-CALCULATERISKSCORE	Code symbol	FU-RISK-SCORING	n/a	risk_score
CODE-CLASSIFYRISK	Code symbol	FU-RISK-SCORING	n/a	risk_score
CODE-CREATEAUDITRECORD	Code symbol	FU-RISK-SCORING	n/a	risk_score
CODE-ENRICHAUDITMETADATA	Code symbol	FU-RISK-SCORING	n/a	risk_score
CODE-HANDLEBANKLINKUNAVAILABLE	Code symbol	FU-PAYMENT-AUTHORIZATION	n/a	payment_
CODE-HANDLEBANKUNAVAILABLE	Code symbol	FU-CHECKOUT	n/a	checkout_
CODE-ISHIGHRISK	Code symbol	FU-RISK-SCORING	n/a	risk_score
CODE-NOTIFYSUPPORT	Code symbol	FU-PAYMENT-AUTHORIZATION	n/a	payment_

ID	Kind	Owner	Description	Source
CODE-PERSISTAUDITRECORD	Code symbol	FU-PAYMENT-AUTHORIZATION	n/a	payment_
CODE-PLACEINREVIEW	Code symbol	FU-PAYMENT-AUTHORIZATION	n/a	payment_
CODE-REQUESTBANKAUTHORIZATION	Code symbol	FU-PAYMENT-AUTHORIZATION	n/a	payment_
CODE-SHOWDECLINEREASON	Code symbol	FU-CHECKOUT	n/a	checkout_
CODE-SHOWRETRYOPTION	Code symbol	FU-CHECKOUT	n/a	checkout_
CODE-SHOWTEMPORARYUNAVAILABLE	Code symbol	FU-CHECKOUT	n/a	checkout_
CODE-STARTSESSION	Code symbol	FU-CHECKOUT	n/a	checkout_
CODE-SUBMITAUTHORIZATIONREQUEST	Code symbol	FU-CHECKOUT	n/a	checkout_

11.5. Verification References

ID	Kind	Status	Description	Source	Canonical
VER-INF-E2E-AUTHORIZED_CHECKOUT_FLOW-8C6597	e2e	not-run	executes authorized checkout flow from session start through payment authorization success	tests/e2e/authorized_checkout_flow.yaml	canonical
VER-INF-INTEGRATION-AUDIT_RECORD_PERSISTENCE_TEST-B333A2	integration	partial	checks audit record persistence includes payment id and computed risk score	test-results/audit-persistence-report.md (+1)	canonical
VER-INF-INTEGRATION-FRAUD_GATE_TEST-12F3BD	integration	pass	validates fraud gate escalation and block decisions before authorization completion	test-results/integration-fraud-gate.json (+1)	canonical
VER-INF-TEST-CONTRACT_CHECKOUT_DECLINE-5D1F26	test	pass	Inferred from test result artifact.	test-results/contract-checkout-decline.json	canonical